

How SONiC Operates Just the Way a Restaurant Does

A simple dining analogy reveals the inner workings of SONiC, a modern open source network OS.



(Image generated by Gemini AI)

SONiC (Software for Open Networking in the Cloud) is an open source, Linux-based network operating system used by hyperscalers and large data centres. It is designed to run on a diverse range of switching hardware from multiple vendors. Its architecture is characterised by a high degree of modularity and containerisation, which are key to its flexibility, reliability, and extensibility.

The core of SONiC's architecture involves a separation of the networking control plane logic from the underlying switch application-specific integrated circuit (ASIC). Key components include:

- **Configuration plane:** Operator and automation inputs (via CLI, config files, or APIs) define the network's desired state, such as routes, VLANs, and telemetry settings, without dictating the hardware implementation details.
- **Orchestration agents (OrchAgent):** This is the central control plane process responsible for reading the desired state from a shared database. It translates the high-level operational intent into low-level, concrete configuration tasks for the hardware layer.
- **Switch Abstraction Interface (SAI):** SAI is a standardised,

vendor-agnostic API. It provides a common interface for programming switch ASICs by defining functions for managing forwarding tables, access control lists (ACLs), and queues. Hardware vendors supply their own SAI implementation to interface with their specific chip.

- **Data plane:** This is the physical forwarding layer where the switch ASIC processes and forwards packets based on the state programmed via SAI. The modular design allows each major function to run as an independent process within its own